



NON-MATERIALIST PHYSICALISM

An experimentally testable conjecture

by David Pearce

"You're nothing but a pack of neurons"
(Francis Crick)

ABSTRACT

Mankind's most successful story of the world, natural science, leaves the existence of consciousness wholly unexplained. The phenomenal binding problem deepens the mystery. Neither classical nor quantum physics seem to allow the binding of distributively processed neuronal micro-experiences into unitary experiential objects apprehended by a unitary phenomenal self. This paper argues that *if* physicalism and the ontological unity of science are to be saved, then we will need to revise our notions of both 1) the intrinsic nature of the physical *and* 2) the quasi-classicality of neurons. In conjunction, these two hypotheses yield a novel, bizarre but experimentally testable prediction of quantum superpositions ("Schrödinger's cat states") of neuronal feature-processors in the CNS at sub-femtosecond timescales. An experimental protocol using *in vitro* neuronal networks is described to confirm or empirically falsify this conjecture via molecular matter-wave interferometry.

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1. Introduction

Natural science promises a complete story of the universe. No "element of reality"⁽¹⁾ should be missing

from the mathematical formalism of physics, i.e. relativistic quantum field theory⁽²⁾ or its more speculative extension, M-theory. On pain of magic, every gross property of the natural world must be theoretically reducible to fundamental physics. The Standard Model in physics is experimentally well tested. Within its conceptual framework, consciousness would seem not only causally impotent but physically impossible. Hence the "explanatory gap"⁽³⁾ and the Hard Problem⁽⁴⁾ of consciousness.

In recent years, a minority of researchers have proposed that the Hard Problem is an artifact of materialist metaphysics. *Contra* Kant⁽⁵⁾, but following Schopenhauer⁽⁶⁾, Bertrand Russell⁽⁷⁾, Grover Maxwell, Michael Lockwood⁽⁸⁾, Galen Strawson⁽⁹⁾, *et al.*, the new idealists conjecture that the phenomenology of one's mind reveals the intrinsic nature of the physical – the elusive "fire" in the equations about which physics is silent. Mathematical physics yields an exhaustive description of the relational-structural properties of the world. This description may ultimately be encoded by the universal wavefunction of post-Everett^a quantum mechanics: our best mathematical description of reality. However, our presupposition that the *intrinsic* character of the physical lacks phenomenal properties is an additional metaphysical assumption. The assumption is hugely plausible, but it's not a scientific discovery. Perhaps most tellingly, the only part of the "fire" in the equations to which one ever enjoys direct access, i.e. one's own consciousness, discloses phenomenal properties that are inconsistent with a materialist ontology. For reasons unexplained, the natural world contains first-person facts. The world supports at least one non-zombie. And natural science gives no reason to believe that one is special.

Untestability cuts both ways. Any conjecture that superpositions of the world's fundamental quantum fields – and, presumably, fundamental macroscopic quantum phenomena such as superconductors or superfluid helium – are intrinsically experiential would seem unfalsifiable too: just speculative metaphysics.

Rather surprisingly, we shall see this isn't the case.

Preliminary Definitions

Both physics and philosophy are jargon-ridden. So let's first define some key concepts.

Both "**consciousness**" and "physical" are contested terms. Accurately if inelegantly, consciousness may be described following Nagel ("What is it like to be a bat?") as the subjective *what-it's-like-ness* of experience. Academic philosophers term such self-intimating "raw feels" "**qualia**" – whether macro-qualia or micro-qualia. The minimum unit of consciousness (or "psychon", so to speak) has been claimed variously to be the entire universe, a person, a sub-personal neural network, an individual neuron, or the most basic entities recognised by quantum physics. In *The Principles of Psychology* (1890), American philosopher and psychologist William James christened these phenomenal simples "primordial **mind-dust**". This paper conjectures that (1) our minds consist of ultra-rapidly decohering neuronal

superpositions in strict accordance with unmodified quantum physics without the mythical "collapse of the wavefunction"; (2) natural selection has harnessed the properties of these neuronal superpositions so our minds run phenomenally bound world-simulations; and (3) predicts that with enough ingenuity the non-classical interference signature of these conscious neuronal superpositions will be independently experimentally detectable (see [6](#) below) to the satisfaction of the most incredulous critic.

The "**physical**" may be contrasted with the supernatural or the abstract. Dualists and epiphenomenalists contrast the physical with the mental. The current absence of any satisfactory "positive" definition of the physical leads many philosophers of science to adopt instead the "*via negativa*". Thus some materialists have sought stipulatively to *define* the physical in terms of an absence of phenomenal experience. Such *a priori* definitions of the nature of the physical are question-begging.

"**Physicalism**" is sometimes treated as the formalistic claim that the natural world is exhaustively described by the equations of physics and their solutions. Beyond these structural-relational properties of matter and energy, the term "physicalism" is *also* often used to make an ontological claim about the *intrinsic* character of whatever the equations describe. This intrinsic character, or metaphysical essence, is typically assumed to be non-phenomenal. "**Strawsonian physicalists**" (*cf.* "Consciousness and Its Place in Nature: Does Physicalism Entail Panpsychism?") and other non-materialist physicalists dispute any such assumption. Traditional **reductive physicalism** proposes that the properties of larger entities are determined by properties of their physical parts. If the wavefunction monism of post-Everett quantum mechanics assumed here is true, then the world does not contain discrete physical parts as understood by classical physics. If contemporary physicalism is true, reductionism is false.

"**Materialism**" is the metaphysical doctrine that the world is made of intrinsically non-phenomenal "stuff". Materialism and physicalism are often treated as cousins and sometimes as mere stylistic variants – with "physicalism" used as a nod to how bosonic fields, for example, are not matter. "**Physicalistic materialism**" is the claim that physical reality is fundamentally *non-experiential* *and* that the natural world is exhaustively described by the equations of physics and their solutions.

"**Panpsychism**" is the doctrine that the world's fundamental physical stuff also has primitive experiential properties. Unlike the physicalistic idealism explored here, panpsychism doesn't claim that the world's fundamental physical stuff *is* experiential. Panpsychism is best treated as a form of property-dualism.

"**Epiphenomenalism**" in philosophy of mind is the view that experience is caused by material states or events in the brain but does not itself cause anything; the causal efficacy of mental agency is an illusion.

For our purposes, "**idealism**" is the ontological claim that reality is fundamentally experiential. This use of the term should be distinguished from Berkeleyan idealism, and

more generally, from subjective idealism, i.e. the doctrine that only mental contents exist: reality is mind-dependent. One potential source of confusion of contemporary scientific idealism with traditional philosophical idealism is the use by inferential realists in the theory of perception of the term "world-simulation". The mind-dependence of one's phenomenal world-simulation, i.e. the quasi-classical world of one's everyday experience, does not entail the idealist claim that the mind-independent physical world is intrinsically experiential in nature – a far bolder conjecture that we nonetheless tentatively defend here.

"Physicalistic idealism" is the non-materialist physicalist claim that reality is fundamentally experiential *and* that the natural world is exhaustively described by the equations of physics and their solutions: more specifically, by the continuous, linear, unitary evolution of the universal wavefunction of post-Everett quantum mechanics. The **"decoherence program"** in contemporary theoretical physics aims to show in a rigorously quantitative manner how quasi-classicality emerges from the unitary Schrödinger dynamics.

"Monism" is the conjecture that reality consists of a single kind of "stuff" – be it material, experiential, spiritual, or whatever. *Wavefunction monism* is the view that the universal wavefunction mathematically represents, exhaustively, all there is in the world. Strictly speaking, wavefunction monism shouldn't be construed as the claim that reality literally consists of a certain function, i.e. a mapping from some mind-wrenchingly immense configuration space to the complex numbers, but rather as the claim that every mathematical property of the wavefunction, except the overall phase, corresponds to some property of physical world.

"Dualism", the conjecture that reality consists of two kinds of "stuff", comes in many flavours: naturalistic and theological; interactionist and non-interactionist; property and ontological. In the modern era, most scientifically literate monists have been materialists. But to describe oneself as both a physicalist *and* a monistic idealist is not the schizophrenic word-salad it sounds at first blush.

"Functionalism" in philosophy of mind is the theory that mental states are constituted solely by their functional role, i.e. by their causal relations to other mental states, perceptual inputs, and behavioural outputs. Functionalism is often associated with the idea of "substrate-neutrality", sometimes misnamed "substrate-independence", i.e. minds can be realised in multiple substrates and at multiple levels of abstraction. However, *micro*-functionalists may dispute substrate-neutrality on the grounds that one or more properties of mind, for example phenomenal binding, functionally implicate the world's quantum-mechanical bedrock from which the quasi-classical worlds of Everett's multiverse emerge. Thus this paper will argue that only successive quantum-coherent neuronal superpositions at naively preposterously short time-scales can explain phenomenal binding. Without phenomenal binding, no functionally adaptive classical world-simulations could exist in the first instance.

The **"binding problem"**⁽¹⁰⁾, also called the "combination problem", refers to the mystery of how the micro-experiences mediated by supposedly discrete and

distributed neuronal edge-detectors, motion-detectors, shape-detectors, colour-detectors, etc., can be "bound" into unitary experiential objects ("**local**" binding) apprehended by a unitary experiential self ("**global**" binding). Neuroelectrode studies using awake, verbally competent human subjects confirm that neuronal micro-experiences exist. Classical neuroscience cannot explain how they could ever be phenomenally bound. As normally posed, the binding problem assumes rather than derives the emergence of classicality.

"**Mereology**" is the theory of the relations between part to whole and the relations between part to part within a whole. Scientifically literate humans find it's natural and convenient to think of particles, macromolecules or neurons as having their own individual wavefunctions by which they can be formally represented. However, the manifest *non*-classicality of phenomenal binding means that in some contexts we must consider describing the entire mind-brain via a single wavefunction. Organic minds are not simply the "mereological sum" of discrete, decohered classical parts. Sentient organic brains are not simply the "mereological sum" of discrete, decohered classical neurons.

"**Quantum field theory**" (QFT) is the formal, mathematico-physical description of the natural world. The world is made up of the states of interacting quantum fields, conventionally non-experiential in character, that take on discrete values. Physicists use mathematical entities known as "**wavefunctions**" to represent quantum states. Wavefunctions may be conceived as representing all the possible configurations of a superposed quantum system. Wavefunction(al)s are complex-valued functionals on the space of field configurations. Wavefunctions in quantum mechanics are sinusoidal functions with an amplitude (a "measure") and

also a phase. The Schrödinger equation $H(t)|\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$ describes the time-evolution of a wavefunction.

"**Coherence**" means that the *phases* of the wavefunction are kept constant between the coherent particles, macromolecules or (hypothetically) neurons, while

"**decoherence**" is the effective loss of ordering of the phase angles between the components of a system in a quantum superposition due to interactions with the environment. Such thermally-induced "dephasing" rapidly leads to the emergence – on a perceptual naive realist story – of classical, i.e. probabilistically *additive*, behaviour in the central nervous system ("**CNS**"), and also the illusory appearance of separate, non-interfering organic macromolecules. Hence the discrete, decohered classical neurons of laboratory microscopy and biology textbooks. Unlike classical physics, quantum mechanics deals with superpositions of probability *amplitudes* rather than of probabilities; hence the *interference*-terms in the probability distribution. Decoherence should be distinguished from *dissipation*, i.e. the loss of energy from a system – a much slower, classical effect. Phase coherence is a quantum phenomenon with no classical analogue. If quantum theory is universally true, then any physical system such as a molecule, neuron, neuronal network or an entire mind-brain exists partly in all its theoretically allowed states, or configuration of its physical properties, *simultaneously* in a "**quantum superposition**"; informally, a "Schrödinger's cat state", a

weighted combination of all possible measurement outcomes. Each state is formally represented by a complex vector [technically a *ray*, or one-dimensional subspace] in Hilbert space. Hilbert space is the generalisation of Euclidean space containing the wavefunctions standing for the possible states of any physical system. Whatever overall state the nervous system is in can be represented as being a superposition of varying amounts of these particular states ("eigenstates") where the amount that each eigenstate contributes to the overall sum is termed a *component*. The "**Schrödinger equation**" is a partial differential equation that describes how the state of a physical system changes with time. The Schrödinger equation acts on the entire probability amplitude, not merely its absolute value. The absolute value of the probability amplitude encodes information about probability densities, so to speak, whereas its phase encodes information about the interference between quantum states. On measurement by an experimenter, the value of the physical quantity in a quantum superposition will naively seem to "collapse" in an irreducibly stochastic manner, with a probability equal to the square of the coefficient of the superposition in the linear combination. If the superposition principle really breaks down in the mind-brain, as traditional Copenhagen positivists still believe, then the central conjecture of this paper is false.

"**Mereological nihilism**", also known as "compositional nihilism", is the philosophical position that objects with proper parts do not exist, whether extended in space or in time. Only basic building blocks (particles, fields, superstrings, branes, information, micro-experiences, quantum superpositions, entangled states, or whatever) without parts exist. Such ontological reductionism is untenable if the mind-brain supports macroscopic quantum coherence in the guise of bound phenomenal states because coherent neuronal superpositions describe *individual* physical states. Coherent superpositions of neuronal feature-detectors cannot be interpreted as classical ensembles of states. Radical ontological reductionism is even more problematic if post-Everett⁽¹¹⁾ quantum mechanics is correct: reality is exhaustively described by the time-evolution of one gigantic universal wavefunction. If such "**wavefunction monism**" is true, then talk of how neuronal superpositions are rapidly "destroyed" is just a linguistic convenience because a looser, heavily-disguised coherence persists within a higher-level Schrödinger equation (or its relativistic generalisation) that subsumes the previously tighter entanglement within a hierarchy of wavefunctions, all ultimately subsumed within the universal wavefunction.

"**Direct realism**", also known as "naive realism", about perception is the pre-scientific view that the mind-brain is directly acquainted with the external world. In contrast, the "**world-simulation model**"⁽¹²⁾ assumed here treats the mind-brain as running a data-driven *simulation* of gross fitness-relevant patterns in the mind-independent environment. As an inferential realist, the world-simulationist is not committed *per se* to any kind of idealist ontology, physicalistic or otherwise. However, s/he will understand phenomenal consciousness as broader in scope compared to the traditional perceptual direct realist. The world-simulationist will also be less confident

than the direct realist that we have any kind of pre-theoretic conceptual handle on the nature of the "physical" beyond the formalism of theoretical physics – and our own phenomenally-bound physical consciousness.

"Classical worlds" are what perceptual direct realists call *the* world. Quantum theory suggests that the multiverse exists in an inconceivably vast cosmological superposition. Yet within our individual perceptual world-simulations, familiar macroscopic objects 1) occupy definite positions (the "preferred basis" problem); 2) don't readily display quantum interference effects; and 3) yield well-defined outcomes when experimentally probed. Cats are either dead or alive, not dead-and-alive. Or as one scientific populariser puts it, "Where Does All the Weirdness Go?" This paper argues that the answer lies under our virtual noses, so to speak – though independent physical proof to silence sceptics will depend on next-generation matter-wave interferometry. Phenomenally-bound classical world-simulations *are the mind-dependent signature of the quantum "weirdness"*. Without the superposition principle, no phenomenally-bound classical world-simulations could exist and nor could minds. In short, we shouldn't imagine superpositions of live-and-dead *cats*, but instead think of superpositions of colour-, shape-, edge- and motion-processing *neurons*. Thanks to natural selection, the *content* of our waking world-simulations typically appears classical; but the *vehicle* of the simulation that our minds run is inescapably quantum. If the world were classical it wouldn't look like anything to anyone.

A **"zombie"**, sometimes called a "philosophical zombie" or "p-zombie" to avoid confusion with its lumbering Hollywood cousins, is a hypothetical organism that is materially and behaviourally identical to humans and other organic sentients but which isn't conscious. Philosophers explore the epistemological question of how each of us can know that s/he isn't surrounded by p-zombies. Yet we face a mystery deeper than the ancient sceptical Problem of Other Minds. *If* our ordinary understanding of the fundamental nature of matter and energy as described by physics is correct, and *if* our neurons are effectively decohered classical objects as suggested by standard neuroscience, then we all ought to be zombies. Following David Chalmers, this is called the **Hard Problem** of consciousness.

Why aren't we P-Zombies? Why aren't we Micro-Experiential Zombies?

A scientifically adequate theory of conscious mind must explain:

- 1) Why consciousness exists at all.
- 2) How consciousness has the causal power to allow intelligent agents to investigate its own nature.
- 3) How consciousness can be phenomenally "bound" in seemingly classically forbidden ways into unitary dynamic objects. In other words, which of the world's information-processing systems are unitary subjects of experience and which are mere aggregates or "zombies"?

4) Why and how consciousness manifests its diverse textures – ranging from phenomenal colours, sounds, tastes and smells, pains and pleasures, the experience of introspecting a thought-episode, feeling pangs of jealousy, hearing an orchestra play, admiring a sunset, to finding a joke amusing. In our mathematico-physical Theory of Everything (TOE), where is the *information* that yields the disparate values of experience?

Finally, any satisfactory scientific theory of consciousness should also offer predictions that are both *novel* and experimentally *falsifiable*.

2. Challenges to Non-Materialist Physicalism.

David Chalmers⁽¹³⁾ identifies two challenges faced by any claim that consciousness discloses the intrinsic nature of the physical:

- a) *the argument from microphysical simplicity.*
- b) *the argument from structural mismatch.*

Let us look at these two challenges in turn.

a) The argument that if physicalistic idealism is true, then "we can expect only a handful of microqualities, corresponding to the handful of fundamental microphysical properties" is intuitively appealing. After all, runs this line of argument, every electron in the world is type-identical to every other electron. Electrons are exceedingly simple. After we've specified the mass, charge and spin of an electron, what else is there to say? An electron "has no hair". Or more technically, after we have given the four quantum numbers that completely describe the electron, namely its principal quantum number(*n*), azimuthal quantum number(*l*), magnetic quantum number(*m*), and spin quantum number(*s*), what else is there left to add?

However, in quantum field theory rather than basic quantum mechanics, there are no particles, only fields and field quanta. What we call "particles" by cosy analogy with classical physics are emergent entities supervenient on the underlying quantum fields. So if instead of a particle-based ontology, the monistic idealist assumes a quantum field-theoretic ontology, then the diverse values of the world's fundamental fields yield the diverse subjective textures of micro-qualia – a vast palette of different qualia-field values. All physical systems, including macroscopic neural networks, are quantum fields. To be sure, in our present ignorance we don't know how to "read off" the diverse values of micro-qualia from superpositions of the diverse values of the different fundamental fields. We lack any kind of cosmic Rosetta stone. But on this physicalistic idealist conjecture, there is no "element of reality" lacking in the quantum field-theoretic formalism that encodes the world's fundamental micro-experiences. Algorithmically compressed into mathematical equations, the information encoding the exact textures of qualia-field values just awaits extraction. For in contemporary physics, fields (or indeed superstrings or branes⁽¹⁴⁾) are defined purely mathematically, even though their experimentally manipulable effects show that the fields are physically real. These fields take a vast range of values ("numbers in space") – with a (conventionally) infinite number of

degrees of freedom. And crudely, on this account "more is different" – micro-experientially different. Critically, these fields aren't classical. Overcoming Chalmers' second challenge to physicalistic idealism (b), i.e. the argument from structural mismatch, turns on recognising that fields in quantum field theory exist in quantum *superpositions* of states. These quantum superpositions may be microscopic, mesoscopic or macroscopic: all are subject to the laws of quantum physics.

b) The argument that the "macrophenomenal structure of my visual field is *prima facie* very different from the macrophysical structure of my brain" seems intuitively obvious too. Yet this intuitive appeal may simply reflect the coarse-grained temporal resolution of our tools for investigating awake/dreaming mind-brains: a resolution of milliseconds not picoseconds, femtoseconds and attoseconds. Appearances of a structural mismatch between neuroscience and phenomenology may be deceptive. There is no experimental evidence for a breakdown of the superposition principle in the mind-brain. What the textbooks call *synchronous* firings of classical neuronal feature-detectors may turn out to be successive quantum-coherent *superpositions* of the relevant neuronal feature-detectors. We won't know whether superposition is masquerading to experimental neuroscience as synchrony until advanced interferometry experimentally settles the issue independently. If empirically confirmed, the detection of such sub-femtosecond neuronal superpositions would render a stunningly beautiful result; the experimental confirmation of what sounds naively like unbridled metaphysical speculation.

Let's use a nonbiological analogy. If physicalistic idealism is true, then the macro-phenomenal structure of superfluid helium presumably consists of a simple, unvarying, long-lived, irreducible macro-experience: a perfect structural match between the formal and subjective properties of the world. Of course, humans will never know what – if anything – it's like to instantiate the wavefunction that describes superfluid helium. But when our experimental apparatus allows probing the CNS at the sub-femtosecond timescales below which e.g. Max Tegmark ("Why the brain is probably not a quantum computer") posits effectively irreversible thermally-induced decoherence, then our classical intuitions may be confounded. On this conjecture, we will find, not random quantum "noise", but instead the structural quantum-coherent physical shadows of the bound macroscopic phenomenal objects of everyday experience – all computationally optimised by hundreds of millions years of evolution to track fitness-relevant patterns in the mind-independent world, i.e. a perfect structural match, not a mismatch, between the phenomenology of consciousness and our canonical representations of the physical. According to the conjecture here explored, training up our neural networks ensures that some neuronal states of the CNS are less prone to thermally-induced decoherence than others. It's these *comparatively* robust experiential-physical states, most notably the perceptual objects of everyday experience, that experimentalists will detect in the CNS when molecular matter-wave interferometry catches up with theory. So when you report, "I can see a chair", and (on the conventional classical story)

synchronous activation of your relevant neuronal feature-detectors occurs, the conjecture will be *falsified* if the subtle non-classical neuronal interference effects typically detected are irrelevant "noise", say a sub-attosecond superposition of the neurons synchronously activated when you see a hippopotamus (etc). In other words, the putative mismatch that Chalmers identifies between the phenomenology of our bound phenomenal minds and the architecture of the brain may turn out to be an artifact of the low temporal resolution of our clumsy tools of investigation.

This is, most certainly, an unintuitive hypothesis. Yet the neurological implausibility of such a fine-grained temporal match should be set against the physical incredibility of the alternatives. From the perspective of natural science, the discovery of a true structural mismatch between physics and phenomenology in the CNS would be *more* astonishing than the previously unsuspected isomorphism between the phenomenal and the physical canvassed here. Such a rupture in the fabric of reality would spell the end of physicalism – an epistemic catastrophe for the unity of science. Unlike his critics, David Chalmers is right to recognise the magnitude of the structural mismatch problem for orthodox materialism and classical panpsychism alike. Chalmers just quits the game too soon. He embraces what must surely count as a counsel of despair: dualism. Monistic physicalism *can* still be saved. Physicalism would be unsalvageable only if the brain is no more than a networked community of discrete, effectively *classical* neurons – or their idealist counterpart, i.e. discrete, effectively classical neuronal "mind-dust" – rather than a succession of macroscopic neuronal superpositions that make up one's everyday phenomenal world. Monistic physicalism *isn't* falsified by a structural mismatch between the three-dimensional space of naive perceptual realism and conscious mind. Monistic physicalism would be falsified only by a structural mismatch between the bound phenomenology of our minds and the fundamental high-dimensional space required by the dynamics of the wavefunction. No such mismatch has ever been experimentally demonstrated to date.

3. Phenomenal Binding is the Hallmark of Mind.

"The only realities are the separate molecules, or at most cells. Their aggregation into a 'brain' is a fiction of popular speech", said William James in *The Principles of Psychology* (1890). The existence of bound phenomenal minds rather than cellular mind-dust suggests that separate molecules and nerve cells are a fiction of classical neuromythology.

Perhaps the greatest cognitive achievement of post-Cambrian⁽¹⁵⁾ central nervous systems has been to solve the binding problem. Without phenomenal binding, members of the animal kingdom wouldn't have minds at all – or classical-seeming world-simulations they could navigate. Over the past half-billion or more years, the mind-brains of unprogrammed organic robots have evolved under the pressure of natural selection to run data-driven, cross-modally matched egocentric world-simulations of their local environment *in almost real time*.

The extraordinary computational power of binding is most vividly illustrated in neurological syndromes where local or global phenomenal binding partially breaks down.

Patients with simultanagnosia⁽¹⁶⁾, for example, can see only one phenomenal object at once. Victims of cerebral akinetopsia⁽¹⁷⁾ are unable to detect motion. People with florid schizophrenia suffer from the disintegration of a unitary self. Even partial loss of phenomenal binding may be intellectually debilitating and behaviourally catastrophic. Neurotypical minds carry off such computational feats with ease. Unfortunately, a neuroscientific explanation is elusive.

By way of context, the phenomenal binding problem is normally posed roughly as follows. How can what neuroscience suggests are distributively neurally-processed edges, colours, shapes, motions (etc) be "bound" into unitary experiential objects populating a unitary experiential field instantiated by a fleetingly unitary self in the neural networks of the CNS? Such phenomenal binding would seem impossible for discrete, membrane-bound, quasi-classical neurons – or quasi-classical "mind-dust" on a physicalistic idealist ontology – separated by c. 3.5 nanometre electrical gap-junctions and 20-40 nanometre chemical synapses. Mere synchronous activation of discrete, decohered classical systems cannot bind – any more than discrete skull-bound minds each undergoing a pinprick causes the emergence of a global mega-mind in agony, or a musical symphony emerges from discrete skull-bound minds each instantiating a musical note. Whether causally connected or otherwise, synchronously activated classical "pixels" of experience remain unglued. Phenomenal mind is not a classical phenomenon. Neither are the pseudo-classical world-simulations run by our waking minds. Chalmers is right on that score. Does quantum mind-binding fare any better?

On the face of it, no. Decoherence is among the fastest processes known to experimental physics. By contrast, we normally assume that states of consciousness somehow arise via neural transmission over a time-scale of milliseconds. Yet unless, implausibly, quantum theory breaks down in the mind-brain – as in so-called "dynamical collapse" or "hidden variables" theories of QM – macroscopic quantum-coherent states implicating such neurologically distributed cellular processes of feature-processing *must* exist in the CNS. What's in question is only their character: noise or signal? Classical neuroscience assumes that these neuronal superpositions are irrelevant to consciousness.

To make our point, let's pose a concrete question. When one apprehends a bound phenomenal object in one's world-simulation, what does it feel like to instantiate successive quantum-coherent macro-superpositions of colour-detector neurons, motion-detector neurons, edge-detector neurons, etc., with each macro-superposition in the sequence lasting what theory suggests must be a femtosecond or less? The obvious answer to the question of what-it-feels-like to instantiate such a sequence of neuronal superpositions is "nothing at all", or perhaps computationally incidental "psychotic noise", because environment-induced decoherence effectively destroys macroscopic neuronal superpositions in the CNS at sub-

femtosecond timescales. Quantum coherence is for all practical purposes irreversibly delocalised into the larger CNS-environment combination though uncontrolled environmental entanglement. On the standard neuroscientific story, our conscious macro-experiences of bound phenomenal objects apprehended by a unitary phenomenal self somehow "arise" instead from patterns of classical, decohered neuronal action potentials synchronously firing over timescales of milliseconds. Yet an answer of "nothing at all" to what-it-feels-like to instantiate a sub-femtosecond neuronal superposition is *not* a possible response for the non-materialist physicalist. For if non-materialist physicalism is true, then phenomenal simples are the world's intrinsic physical properties – the "fire" in the equations of quantum field theory. A fleeting macroscopic neuronal superposition is just such a phenomenal simple: it's not an aggregate or classical ensemble of anything more primitive. Classical glue cannot bind; quantum-coherent glue can't do anything else; thermally-induced decoherence in the CNS explains just how rapidly our fragile minds become unstuck. Thus decoherence can be viewed as a progressive phenomenal *unbinding* – or in other words, effective dephasing is a solution to the phenomenal *unbinding* problem. The universal wavefunction is not a mind.

The alternative to such a perfect structural match hypothesis is equally stark. Unless we abandon the conceptual framework of physicalism, then mere synchronous⁽¹⁸⁾ neuronal firings *cannot* phenomenally bind purely classical neurons or neuronal "mind-dust" into cross-modally matched phenomenal objects, or a spatio-temporally unitary perceptual field, or a transiently unitary phenomenal self. Mere synchronous neuronal firings cannot bind any more than, say, synchronous activation and reciprocal electromagnetic communications could phenomenally bind a community of skull-bound American minds. It's not that we can disprove Eric Schwitzgebel's claim that "If Materialism Is True, the United States Is Probably Conscious"⁽¹⁹⁾. Rather, the emergence of such a unitary pan-continental subject of experience would be unexplained and inexplicable – a miracle in all but name. Such spooky strong ontological emergence would violate physicalism in a most spectacular way. By the same token, *if* our 86 billion odd neurons always behaved as essentially classical systems, as they do in a dreamless sleep or coma, then the emergence of a unitary pan-cerebral subject of experience, a "person", would be unexplained and inexplicable as well. Such spooky strong emergence would violate physicalism too.

Of course, the difference between the USA and the mind-brain is that – unlike a hypothetical pan-continental subject of experience – it's hard to treat the existence of one's own conscious mind as simply a conjecture. Rather the existence of one's bound conscious mind is what needs to be explained – unless you happen to be a philosophical zombie.

That said, eliminative materialists like Daniel Dennett⁽²⁰⁾ are right to recognise that qualia (raw phenomenal experiences) are impossible within a materialist ontology. More particularly, eliminative materialists are right to

recognise that the existence of qualia in the brain – as understood by classical materialistic neuroscience – is a physical impossibility, whether the existence of phenomenal symphonies, chairs, tables, mountains, or the whole panoply of lived experience. Yet this alleged impossibility derives from a combination of our classical *misrepresentations* of the mind-brain; our temporally coarse-grained observations of other central nervous systems; and our quasi-hardwired perceptual naive realism with the crude materialist ontology it spawns. We're not entitled to infer that humans must be insentient zombies on the grounds that our materialist ontology can find no naturalistic place for our sentience. The eliminative materialist who forgoes anaesthesia before surgery has yet to be born.

An obvious counterargument to such a (presently hypothetical) perfect fine-grained match between the phenomenology of our minds and the physical structure of the CNS is that we perceive our surroundings with a time-lag of scores of milliseconds. Such a time-lag is orders of magnitude too long for ultra-rapidly thermally "destroyed" (i.e. lost to the environment in a thermodynamically irreversible way) quantum-coherent neuronal superpositions to be computationally relevant to perception.

This objection presupposes an untenable perceptual naive realism in which we directly "see" the mind-independent world – the same misconceived perceptual naive realism according to which a neurosurgeon directly "sees" the cheesy wet nervous tissue constituting the mind-brain of an anaesthetised patient lying on his operating table prior to surgery.

Such perceptual naive realism may be compared with Bertrand Russell's apt reminder that one never "sees" anything but the inside of one's own head. Not even a neurosurgeon in the operating theatre. According to our contrasting world-simulation model, the role of the local mind-independent environment is essentially to *select* quantum-coherent superpositions of the awake mind/brain via optic and other nerve inputs with an evolutionarily minimised time-lag.

If Max Tegmark's calculations⁽²¹⁾ – as distinct from his conclusions – are approximately correct, then our world-simulations must run at anything from around 10^{13} quantum-coherent neuronal "frames" per second up to a frame-rate of up to 10^{20+} quantum-coherent neuronal "frames" per second. The fitness-relevant environmental patterns that they track in waking states lag behind their neural counterparts by a hundred or more milliseconds. In that sense, we always "live in the past"; but our waking world-simulations run in near enough to real time for organic robots to behave flexibly and adaptively in an inhospitable environment.

A suggestive analogy here might be the persistence of vision undergone by organic minds watching a movie run at 24-frames-per-second. Each composite frame of the movie can be rich, diverse and multifaceted, despite the lack of perceptible individuality or any "gappiness" to our minds when the frame-sequence is run. The film would seem the same if it were a notional $10^{15} \times 24$ -frames-

per-second movie. However, the inner theatre metaphor of mind can also mislead. This is because such a metaphor seems to generate an infinite regress of homunculi. How is the inner spectator supposed to view the internal scene if not by means of another inner spectator in turn? And so forth. In reality, our minds partly *instantiate* the virtual world-simulations they run. All analogies break down somewhere; the Cartesian theatre is no exception. Note that the phenomenal unity of perception at issue here is what philosophers call "synchronic" unity. No claim is being made about "diachronic" unity, the fictitious temporal persistence of an enduring metaphysical ego. Such enduring personal identity is fundamental to our conceptual scheme. Yet persisting selves are impossible to reconcile with a physicalistic world-picture, not least with the rapid metabolic turnover of one's constituents – or with the existence of one's 10^{100} near-identical namesakes that post-Everett quantum mechanics implies partially decohered ("split") since the start of this sentence⁽²²⁾. For expository convenience, the narrative fiction of enduring personal identity will here be retained. In principle, however, each ultra-thin "slice" or quantum-coherent frame of episodic self could be labelled with its own numerical subscript.

A more robust *a priori* objection to quantum mind hypotheses of phenomenal binding might run as follows. No, says the traditional materialist and coarse-grained functionalist, we don't yet understand how consciousness arises from patterns of neuronal firings in the brain. But as reductive physicalists, we *shouldn't* be surprised at the structural mismatch between the phenomenology of bound phenomenal objects and the micro-structure of the brain, *per se*, any more than we should be surprised at the structural mismatch between video game characters and the program code running on the classical computer processor that executes them. There's no need to invoke quantum "woo" when well-understood classical physics and learning algorithms work just fine. Indeed, the same point could be made of a massively classically parallel, "sub-symbolic" connectionist⁽²³⁾ information-processing system that lacks the transparent and projectable⁽²⁴⁾ representations of a classical serial computer. Connectionist systems are sometimes called "neural networks" in recognition of their closer gross architectural resemblance to the mind-brain than a programmable serial digital computer.

Unfortunately, this argument doesn't work either. For sure, when speaking colloquially as though perceptual direct realism were true, we can talk about seeing visually bound video-game characters battling their way across one's computer monitor. And yes, these classical computer-created video-game characters are generated via well-understood, classical computations. No need for quantum "woo" here. However, the manifest phenomenal binding is done entirely by – and is entirely internal to – the sentient organic minds playing the video game: it's internal to the phenomenal world-simulations of the game's players. Such binding is not a property of anything internal to the mind-independent computer display unit. Video game characters lack true ontological integrity: they are not unitary subjects of experience running phenomenally bound world-simulations of their

own. All that exists in the mind-independent world are thousands (or millions) of effectively discrete pixels on a monitor screen. Whether our fundamental ontology of the natural world is materialist, panpsychist or idealist in character, these effectively classical pixels do not generate phenomenally unitary subjects of experience that sentient minds engage in combat. Rather, their programmed patterns are part of the distal causal chain that culminates in the minds of organic sentients as video-game characters, i.e. their patterns on a monitor causally covary with the phenomenal game avatars populating the phenomenal gadgets of our phenomenal world-simulations. Effectively, there are no bound phenomena in our personal computers to be matched or mismatched.

In fairness, the insentience of digital zombies has been challenged (*cf.* "This guy thinks killing video game characters is immoral."⁽²⁵⁾). Quantum mind-binding theory as defended here vindicates sceptical common-sense.

So what would an exact structural match between bound experiential objects and the superposition of state vectors of neuronal edge-detectors, motion-detectors and colour-detectors (etc) over ultra-Tegmarkian time-frames entail? In "Are Perceptual Fields Quantum Fields?"⁽²⁶⁾, Brian Flanagan aptly cites Dirac:

"When a state is formed by the superposition of two other states, it will have properties that are in some vague way intermediate between those of the original states and that approach more or less closely to those of either of them according to the greater or less 'weight' attached to this state in the superposition process. The new state is completely defined by the two original states when their relative weights in the superposition process are known, together with a certain phase difference, the exact meaning of weights and phases being provided in the general case by the mathematical theory. (Dirac, PAM. *The Principles of Quantum Mechanics*. Oxford, 1958)

Of course, Dirac wasn't assuming quantum-coherent superpositions of *qualia*-fields, but rather quantum-coherent superpositions of fields of *non*-phenomenal we-know-not-what. Moreover, Dirac was writing about quantum microphysics, not short-lived superpositions of mesoscopic and macroscopic phenomenal objects in warm, wet organic brains. Yet what if our traditional insistence on a non-phenomenal metaphysical essence to our field-theoretic ontology is dropped? Quantum field theory is no more inherently about fields of *insentience* than Maxwell's Theory of Electromagnetism is inherently about the properties of luminiferous aether. (*cf.* Heinrich Hertz's terse observation, "Maxwell's Theory is Maxwell's equations.") Neither theoretical physics nor the phenomenology of mind give any comfort to the idea that the superposition principle really breaks down in the CNS.

4. Can Physicalism be Saved?

Picoseconds are of an unimaginably, mind-wrenchingly long duration compared to the fundamental Planck scale of around 10^{-43} seconds - over thirty orders of magnitude more protracted. Femtosecond, attosecond and even pan-cerebral zeptosecond rates of environment-induced decoherence are still staggeringly long-drawn-out durations once we leave the everyday intuitions of folk-chronology behind. Nonetheless, most neuroscientists would confidently predict that invoking an exact phenomenal-physical structural match at Tegmarkian temporal resolutions to solve the phenomenal binding problem is not just (potentially) falsifiable but false. All we'll discover via interferometry in the warm and wet CNS at such time-scales is an uninteresting, functionally irrelevant and effectively random thermally-induced "noise", not the structural shadows of bound phenomenal objects. After all, picoseconds are seven or eight orders of magnitude shorter than the widely accepted time-frame over which electrochemical neuronal firings cause consciousness to "emerge", inexplicably, in our central neural networks. And pan-cerebral quantum-coherent neuronal superpositions can credibly subsist only for sub-attosecond time-frames before the well-defined phase relations between the components of the superposition are lost, i.e. extended to the extra-neuronal environment in a thermodynamically irreversible fashion.

Again, perhaps orthodoxy is correct. At issue here is a scientifically falsifiable conjecture, not a purely "philosophical" claim. Yet if folk neurochronology is vindicated, then the prospects for physicalism and the ontological unity of science are bleak, or simply non-existent. If folk neurochronology is vindicated, something ontologically irreducible is present in the world and missing from the formalism of physics. The spectre of "strong" emergence rears its head – or worse, dualism, whether avowedly "naturalistic" dualism or otherwise. True, materialists and epiphenomenalists don't face the binding problem in quite the same way as the physicalistic idealist. Instead, bound phenomenal objects can simply "emerge" in the brain, like Athena sprung fully formed from the head of Zeus.

The ontological floodgates are opened.

5. What Is It Like To Be Schrödinger's Cat?

So let us provisionally suppose, in defiance of orthodox neuroscience but in conformity with the formalism of unmodified quantum physics, that our prediction of a perfect physical-phenomenal structural match in the CNS turns out to be correct, confounding Chalmers and thereby lending experimental weight to an idealist ontology of monistic physicalism. Rather than embrace epiphenomenalism or Chalmersian dualism, we may on this story transpose the entire mathematical machinery of modern physics to describe an idealist ontology. According to this proposal, sentient beings are wavefunctions in configuration space - fields of phenomenally bound subjective experience whose exact textures are expressed by the values of two numbers, the amplitude and the phase, specified at every point in the universe's configuration space: physicalistic idealism. Every mathematical property of the wavefunction (except the overall phase) corresponds to some subjective

property of the physical world.

Suspending disbelief, what would be the pay-off if this conjecture is true?

Let's return to the criteria that must be satisfied by a scientifically adequate theory of conscious mind.

1) *Why consciousness exists at all.*

This question is best recast as "why is there something rather nothing?" Or more poetically, Hawking's "What is it that breathes fire into the equations and makes a universe for them to describe?" Mysteries should not be multiplied beyond necessity. By positing a non-phenomenal "fire" in the equations, and then hand-waving on how such non-phenomenal stuff might notionally be transmuted into something phenomenal yet still (somehow) material, avowed materialists build a speculative dualistic ontology into their conceptual framework right from the outset. Compare the Catholic doctrine of transubstantiation. The bread and wine used in the sacrament of the Eucharist literally become the body and blood of Christ while all of their features accessible to the senses remain unchanged. In both cases, we confront "a mystery surpassing all understanding".

There *is* one fundamental mystery. Why does anything exist at all?

When investigating why the enigmatic "fire" of physical consciousness exists, perhaps the fundamental *qualia-fields* of a quantum vacuum, perhaps we might explore some kind of zero ontology as the ultimate logico-physical principle underlying reality, with the field values of the world's hypothetical fundamental microqualia "cancelling out" to zero in a multiverse of net zero information. Within this research program, Max Tegmark's "Does the universe in fact contain almost no information?"⁽²⁷⁾ might have considered whether the quantum Library of Babel - our Everettian multiverse? - contains *any* information. For in the absence of a preferred basis, the state vector of Everett's multiverse doesn't *per se* contain any information⁽²⁸⁾. If so, "A theory that explains everything explains nothing" isn't the witty but shallow quip one might assume. No canonically preferred bases of Hilbert space could exist without violating a zero ontology. The mathematical structure of quantum theory allows indefinitely many ways [conventionally, infinitely many ways] to decompose the quantum state of the multiverse into a superposition of orthogonal states. This leads to another question. Is Wigner's "Unreasonable Effectiveness of Mathematics in the Natural Sciences" explained by the need to conserve an informationless zero ontology - the conservation law that forbids

substantive existence? Are the superposition principle and a zero ontology one-and-the-same?

Such difficult questions are beyond the scope of this paper. Proposing that the superposition principle of QM explains both the properties of our minds and why anything exists at all sounds preposterous. But the idea that we may understand either mind or quantum theory *without* understanding why anything at all exists may be naive.

2) How consciousness exerts the causal power to allow intelligent agents to investigate its nature.

According to idealistic physicalism, consciousness has causal efficacy. For instance, a sentient agent removes its hand from the flame because the burning sensation feels agonisingly hot. Unusually, common-sense is actually correct. For sure, in many contexts, for example all programs executed on a classical digital computer, the particular micro-textures of experience constitutive of its phenomenally-unbound physical circuitry are logically and computationally irrelevant to the execution of a program. The particular micro-textures of experience are mere implementation details. But if physicalistic idealism is true, then strictly speaking *all* consciousness, and *only* consciousness, exerts causal power, effectively mediated by what we normally recognise as the four forces of nature, or perhaps, ultimately, the vibration modes of higher-dimensional branes of M-theory. Only the physical has causal efficacy; and consciousness discloses the intrinsic nature of the physical.

Without such causal power, not merely would intelligent agents be unable to investigate consciousness: we wouldn't have grounds for alluding to the existence of consciousness in the first instance. By way of distinction, epiphenomenalists want to claim, presumably, that they have rational *grounds* for believing epiphenomenalism is true - that epiphenomena really are causally impotent. Yet it's unfathomable, to say the least, how such grounds can be stated without implicitly acknowledging a causal role for the epiphenomena that the claim repudiates.

Likewise, on pain of inconsistency, the materialist can't simultaneously assert - as Stephen Hawking⁽²⁹⁾ does most famously in *A Brief History of Time* - that we have no idea of the character of the "fire" in the equations and yet also dispute that its essence could be phenomenal experience. No doubt "fire" consisting of a non-phenomenal *je ne sais quoi* is a plausible speculation. Yet the claim itself borders on the metaphysical. How does the materialist propose to test his conjecture?

We are also now in a position to answer a commonly posed thought-experiment. Materialists wrestling with their Hard Problem of consciousness sometimes wonder why we don't live in a world physically type-identical to our world but populated instead by insentient zombies. Yet if consciousness discloses the intrinsic nature of the physical, and if quantum-coherent phenomenal binding is the hallmark of mind, then a non-sentient world physically type-identical to our world is logically impossible. A possible world can't simultaneously be physically identical and physically non-identical to our world.

To spike some guns, physicalistic idealism isn't a license for free will, human dignity, animism, New Age mysticism, quantum healing, a reconnection with the timeless wisdom of the ancients, or anything warm and fuzzy. Nor does it invoke quantum mechanical "hidden variables". Nor does it claim that "consciousness collapses the wavefunction". Nor is it a variant of Berkeleyan idealism or the philosophical speculations of the German idealists - though Kant's "transcendental unity of apperception" foreshadowed the global binding problem. Nor is it a sceptical hypothesis. Thus the physicalistic idealist believes that the mind-independent world existed long before the evolution of bound phenomenal minds in biological organisms. Across the cosmos, the mathematical straitjacket of relativistic quantum field theory is as tight as ever. Physics - or rather tomorrow's ideal physics beyond the energy range of the Standard Model - is causally closed and complete. But within this naturalistic categorical framework, physics is not assumed to be about some essentially non-phenomenal metaphysical "stuff" or unknowable "fire" beyond the reach of scientific investigation. In more Kantian terminology, consciousness is here conjectured to be the noumenal physical essence of the world, the *Ding an sich* ("thing-in-itself") that Kant assumed would forever be unknown and unknowable. Physicalistic idealism turns Kant on his head⁽⁸⁾. The noumenal world is all one can ever know, or at least a tiny part of it, other than by inference and conjecture. And the phenomenology of even this sliver of direct knowledge is theoretically contaminated; Wilfrid Sellars called the realm of pure non-inferential experience the Myth of the Given⁽³⁰⁾.

3) How consciousness can be phenomenally "bound" in seemingly classically forbidden ways.

Physicalistic idealism is not *animism* or *vitalism*. Its advocates no more believe that a rock is a unified subject of experience than

does, say, an eliminative materialist like Daniel Dennett. In common with every other naturalistic theory, the physicalistic idealist still has a lot of work to do in order to show how a bunch of ostensibly discrete quasi-classical nerve cells (or "mind-dust") can generate bound phenomenal objects or a unitary phenomenal self.

Two key questions arise in tackling the classically insoluble binding problem / combination problem.

- a) Is macroscopic quantum coherence in the CNS a physically real phenomenon?
- b) If so, is the phenomenon long-lived enough to do any computationally and/or experientially useful work - as distinct from being functionally incidental neuronal "noise"?

If quantum mechanics is complete, then the answer to the first question is "yes", albeit over what are, intuitively, vanishingly short durations. However, the existence of macroscopic quantum coherence in the CNS does not, of itself, make the mind-brain a quantum computer any more than the quantum-mechanical properties of silicon (etc) semiconductors make one's desktop PC a quantum computer. One's phenomenal mind and its world-simulation functions as a quantum computer only if what - naively and classically - we describe as the synchronous firings of classically parallel neuronal cellular feature-detectors (edges, colours, shapes, motions, vertices, etc) briefly support a unitary experiential object: the wavefunction of an intelligent, information-processing experiential agent.

This absence of individual neuronal identity in what would otherwise be - as in a dreamless sleep - effectively classical neurons / mind-dust presumably occurs with an ultra-fast "refresh rate" - where "ultra-fast" alludes to our everyday chronological intuitions rather than Planck-scale physics. Within any given sequence of mental life, dropped and mangled frames aren't noticed as such because they aren't explicitly represented in other individual frames. On this story, the molecular structures of our explicit "memories" consolidate only on a much coarser-grained timescale - ranging from hundreds of milliseconds to minutes, hours, days, and in extreme cases, a hundred years or more. Quantum mind-binding isn't a replacement for connectionist neuroscience or its temporally coarse-grained learning algorithms; rather, it's the bedrock.

So what is conscious? Conversely, what's a micro-experiential zombie?

On this touchstone of sentience - i.e. quantum coherence as the physical signature of phenomenal binding - macroscopic

quantum fluids; SQUIDs (Superconducting QUantum Interference Devices); organic mind-brains while not dephased in a coma or dreamless sleep; and perhaps futuristic nonbiological quantum computers are unitary experiential subjects.

Conversely, if effective classicality is the hallmark of the zombie, then serial digital computers, classically parallel connectionist systems, classical dynamical systems, rocks and mountains, the population of the USA, and cellulose-cell-wall-bound plants, etc., are not subjects of experience. Rather they are just decohered aggregates, in effect, composed of phenomenal simples.

Perhaps contrast neuroscientist Giulio Tononi's Integrated information theory⁽³¹⁾ in which consciousness is a function of informational complexity.

Currently, if the quantum mind-binding hypothesis sketched here is true, the largest quantum supercomputer in the world belongs to the sperm whale: a mind around five times heavier than its human counterpart. The human cerebral cortex is 2–4 mm thick; but actually we have to take a four-dimensional approach, or more ambitiously, a finite⁽³²⁾-dimensional Hilbert-space approach, and imagine our minds as 10^{100} or so quasi-classical Everett branches "jostling" each other before becoming irreversibly "split", i.e. effectively decohering and losing their well-defined phase coherence to the extra-cerebral environment. Intuitive plausibility is *not* the hallmark of a scientifically adequate theory of consciousness. Experiment, not philosophy or armchair physics, is the key.

4) Why and how consciousness has its diverse textures – ranging from phenomenal colours, sounds, tastes and smells to pains and pleasures to the experience of introspecting a thought-episode, understanding a text, or finding a joke funny.

If idealistic physicalism is true, then the solutions to the field-theoretic equations of physics mathematically encode the textures and interdependencies of micro-experiences. The amplitude and phase of one's wavefunction yield the exact values of all one's experiences. On this conjecture, there are no hidden parameters or missing variables that the existing quantum-mechanical formalism omits. Quantum mechanics is indeed closed and complete – or more strictly, it will be closed and complete when it subsumes gravity. Hence the spectre of causal over-determination, epiphenomenalism or even dualism in theory of mind is lifted. Another kind of dualism, the

spurious divide between the classical macro-world and the quantum micro-world, evaporates too. The appearance of phenomenally bound classical objects in a classical world is a derived quantum effect, not a brute unexplained fact that a classical materialist ontology can't accommodate.

Finally, any satisfactory theory should offer predictions that are novel, precise, replicable and robustly falsifiable.

Untestable claims may be scientific *if* they are entailed by a conjecture that generates novel, precise, and non-trivial predictions that can be empirically tested. Physicalistic idealism is radically conservative insofar as it does not propose any modification or supplementation of the existing, realistically interpreted, quantum-field-theoretic formalism. *Contra* Penrose and Hameroff's "Orchestrated objective reduction" (Orch-OR⁽³³⁾) model, for example, there is no evidence that the unitary dynamics of standard quantum mechanics breaks down in the central nervous system or anywhere else. Yet – without proposing any new physical law(s) – physicalistic idealism also predicts the existence of an empirically investigable phenomenon that few researchers now credit. Namely, our everyday classical world-simulations are underpinned at sub-femtosecond timescales by macroscopic quantum-coherent physical states of the CNS. One's phenomenally bound quasi-classical virtual world is "what a natural quantum computer feels like from the inside", so to speak. Familiar classical worlds of phenomenally bound objects obeying Newtonian laws of motion and gravity within one's perceptual field are an entirely quantum-mechanical phenomenon. Classical phenomenal macroworlds would be impossible without successive neuronal superpositions of distributed feature-processors to underpin their existence.

What Max Tegmark treats as a *reductio ad absurdum* of quantum mind is treated instead as a falsifiable empirical prediction. Nature got there first; and natural selection got to work.

Of course, if a classically-minded critic is convinced *a priori* that macroscopic quantum-coherent neuronal superpositions of sub-femtosecond duration are of no more computational or phenomenal relevance to explaining consciousness than, say, the detection of evanescent quantum superpositions of, say, the pawns and the queen during a game of chess, or random thermal noise in a classical CPU executing a program on one's PC (etc), then such a critic will not waste time independently setting up the exceedingly delicate experiments

necessary to detect the missing physical signature of phenomenal binding. They might simply say, noise is noise. Collisional decoherence, dephasing due to inertial forces and vibrations, and above all thermal decoherence are all formidable obstacles to detecting the indirect signature of neuronal superpositions even with the tools of next-generation interferometry.

Such a cavalier dismissal of the only way to save physicalism and the ontological unity of science may prove premature. We will now set out the protocol for an experiment to test the naively absurd conjecture that binding-by-synchrony is really binding-by-superposition.

6. Schrödinger's Neurons: the Experimental Protocol.

In vivo experiments using live human subjects or cats are impossible for the foreseeable future. However, cultured *in vitro* neuronal networks should suffice. First, "train up" a multi-layer neuronal network with a suitable input-output device to recognise a variety of externally presented inputs. Then, identify in turn the distributed neuronal feature-processors implicated in diverse object recognition on a standard, classically parallel connectionist account, i.e. "local" phenomenal binding. Routine neural scanning can pick out what we would naively describe as the synchronously activated distributed neuronal feature-processors elicited by any given stimulus, i.e. textbook connectionist neuroscience but using real neurons rather than tendentiously named "artificial neural networks" and their statistical learning algorithms.

Next comes the fiendishly hard part – feasible in principle, but an experimental challenge still beyond the reach of contemporary molecular matter-wave interferometry. Instead of detecting the fleeting non-classical interference patterns of "nonsense" neuronal superpositions, the conjecture predicts that we'll discover the interference signature of sub-femtosecond macro-superpositions that robustly implicate *exactly the same neuronal feature-processors of the synchronously activated neurons that the classical neuroscience story reports are activated in the trained-up neuronal network when object-recognition occurs*. On any classical account of mind, such an experimental outcome, i.e. a perfect structural match, is either physically impossible or vanishingly improbable.

The best-known physically demonstrable manifestation of quantum-coherent superpositions is the interference peaks from an electron wave in a double-slit experiment⁽³⁴⁾. Currently, matter-wave interferometry can detect "mesoscopic" superpositions of fullerenes⁽³⁵⁾ in the guise of observable de Broglie wave interference of C₆₀ and C₇₀ molecules following passage through a diffraction grating. Experimental superpositions of viruses⁽³⁶⁾ and tardigrades ("water bears") are planned. Detecting the interference patterns of neuronal superpositions with their hugely more numerous excited internal degrees of freedom will be much more

challenging because – unlike fullerenes or viruses – functioning neuronal networks can't be steeply cooled down to mitigate the effects of thermally-induced decoherence. In neuronal networks, ion-ion scattering, ion-water collisions, and long-range Coulomb interactions from nearby ions all contribute to rapid decoherence times; but thermally-induced decoherence is even harder experimentally to control than collisional decoherence⁽³⁷⁾.

However, we may assume tomorrow's experimentalists will rise to the challenge. Let's review the possible outcomes. What will experiments detect when molecular matter-wave interferometry can probe the sub-femtosecond timescales over which theory predicts neuronal superpositions should exist?

1) a) no interference effects, or at least some collapse-like deviation from the unitary Schrödinger dynamics, i.e. the superposition principle breaks down in artificial neuronal networks and thus presumably in the CNS. This negative outcome is what Penrose and Hameroff⁽³⁸⁾; Ghirardi, Rimini and Weber (GRW)⁽³⁹⁾; and other dynamical collapse theorists would predict.

or

b) the telltale non-classical interference signature that the unitary dynamics predicts. *IF* b) is the case, then will the sub-femtosecond neuronal superpositions detected be:

2) a) functionally irrelevant psychotic noise, of no more relevance to the orderly phenomenology of our bound phenomenal minds than, say, fleeting sub-femtosecond superpositions of miscellaneous pawns to the gameplay in a chess match? The Chalmersian "structural mismatch" claim is vindicated.

or

b) a perfect structural match that implicates all and only the synchronously firing feature-mediating neurons that orthodox neuroscience reveals are activated when individual phenomenally bound objects are perceived?

Our femto-mind binding conjecture predicts (b) in both cases.

Some comments are in order here.

First, a good experiment should be "clean" and conceptually simple – its outcome decisive to sceptics and hostile critics, not just to the satisfaction of the conjecture's proponents. No scope should exist for fudging, *ad hoc* escape-clauses or adding epicycles. By this criterion, the experiment outlined here is decisive. A critic of quantum mind will be unfazed by such professions of epistemic virtue: by analogy, building a perpetual-motion machine would be a clean, elegant and definitive refutation of the second law of thermodynamics, too; it's not going to happen. Less fancifully, an example of an "unclean" experiment is the discovery of quantum vibrations in microtubules inside brain neurons as a test of the Hameroff-Penrose Orch-OR theory of mind. Their discovery, though intriguing, will not persuade critics that

modified quantum theory makes Gödel-unprovable results provable by human mathematicians.

Second, strictly speaking, it's not necessary to assume that the superposition principle of QM is universal. Maybe spontaneous localisation kicks in at scales larger than the mesoscopic and modestly macroscopic dimensions of organic mind-brains. Such a breakdown would be physically unmotivated. No departure from the Schrödinger dynamics has ever been detected. But the experimental demonstration of neuronal superpositions won't rule it out.

Third, we have avoided fascinating but incidental speculation about e.g. the properties of liquid water as a unique quantum fluid, dipoles forming superposed resonance rings in helical pathways in microtubule lattices⁽⁴⁰⁾, and so forth. For the existence of neuronal superpositions implicating previously naively identified phenomenal feature-mediating nerve cells is a *generic* prediction of any conjecture that invokes coherent superpositions of neuronal feature-processors as the explanation of phenomenal binding. The conjecture – and its confirmation or falsification via matter-wave interferometry – is insensitive to the details of its molecular implementation. Darwin needed Mendel. The ubiquitous selection pressure of Zurek's "quantum Darwinism" applied to the CNS awaits Mendel's counterpart.

Fourth, demonstration of this exceedingly subtle physical interference effect – *if* experimentally confirmed – is not remotely the only reason for believing that organic minds are quantum computers, or that experience discloses the intrinsic nature of the physical. The most striking reason lies in front of our virtual eyes and under our virtual noses, so to speak. But the existence of phenomenal binding is a retrodiction, "old evidence", not a novel prediction. Any claim that armchair philosophising can establish that the mind is a quantum computer will be given short shrift by critics – even if the claim happens to be true. This *in vitro* interferometry experiment is pitched at quantum mind's most implacable foes.

7. Femto-Mind meets Quantum Darwinism.

"I still recall vividly the shock I experienced on first encountering this multiworld concept. The idea of 10^{100} slightly imperfect copies of oneself all constantly splitting into further copies, which ultimately become unrecognizable, is not easy to reconcile with common sense. Here is schizophrenia with a vengeance."⁽⁴¹⁾
(Bryce DeWitt)

If DeWitt's notorious misreading of Everett were true, then we'd be (at most) micro-experiential zombies in all life-supporting branches of the universal wavefunction. Unified subjects of experience and phenomenal binding would be impossible. We'd know nothing of one branch, let alone the googols of others. However, DeWitt was mistaken; there is only one world – the multiverse – and its decohering branches never completely separate. DeWitt's remark nonetheless offers a clue to meeting what might seem a decisive objection to a quantum mind account of phenomenal binding. How could selection

pressure operate over a timescale of femtoseconds, attoseconds or less? The answer is that whereas selection pressure can't act on proliferating worlds, it *can* act on proliferating, decohering world-simulations. In order to understand our minds and the world-simulations they run, Zurek's "quantum Darwinism"⁽⁴²⁾ must be applied to the CNS. Here we have a Darwinian selection-mechanism of unimaginable power: ubiquitous, unremitting and temporally fine-grained. Who will play Mendel to Zurek's Darwin is unknown. These cryptic remarks will now be amplified.

How could non-psychotic phenomenal binding of distributed neuronal feature-processors have evolved? The generation by vertebrate minds of cross-modally matched virtual worlds in almost real time is prodigiously computationally powerful and genetically adaptive. Mere patterns of Jamesian "mind-dust" couldn't *act*. Connectionist neuroscience describes at a coarse-grained level how individual perceptions are represented by shifting coalitions of resting/firing patterns of membrane-bound neuronal feature-processors using different learning algorithms. Yet *if* the phenomenology of virtual world-making ultimately depends on sub-femtosecond quantum coherence, then the *evolution* of non-psychotic phenomenal binding would naively seem impossible. Decoherence, i.e. the rapid effective loss of ordering of the relative phases of complex amplitudes of neuronal superpositions to the environment, is a powerful, omnipresent and seemingly *uncontrollable* effect in the warm, wet CNS.

But we needn't turn to drink or dualism – yet. *If* a femtomind-binding conjecture is correct, and *if* the unitary dynamics of QM doesn't break down in the human mind-brain, then a qualitative answer to the evolutionary enigma of phenomenal binding can be given within the conceptual framework articulated by one of the pioneers of the decoherence program in post-Everett quantum mechanics, Wojciech Zurek. The decoherence program outlines the Darwinian⁽⁴³⁾ process responsible for the emergence of quasi-classical reality from its quantum substrate within Everett's multiverse. *If* a femtomind-binding conjecture is correct, then an analogous Darwinian process of replication, variations amongst the copies, and differential survival of the copies is responsible for the emergence of the quasi-classical phenomenal worlds forming our minds from their quantum substrate in the CNS. Crudely, some superpositions are fitter than others. In order for an ecologically credible quantum mind-binding conjecture to be viable, all that is needed for selection pressure to get to work is the slightest heritable predisposition to the tiniest of transmissible resistance to collisional and thermally-induced decoherence of non-psychotically bound phenomenal neuronal superpositions in even the humblest of cephalic ganglia. All organisms capable of neuronal world-modelling evolve and adapt to their environment by an iterative process. This iterative process may be treated as an evolutionary algorithm that searches the fitness landscape for the locally and globally bound phenomenal states of mind – quantum-coherent neuronal superpositions – that are best adapted to their local surroundings. Thus a Darwinian process of variation and differential selection of informational superpositions

plays out as the fittest phenomenally bound variants are retained and passed on to their offspring. It's worth stressing again: *contra* de Witt's colourful quote above, there is only one multiverse; interference effects between Everett branches that have effectively decohered ("split") never wholly disappear. Within the universal wavefunction, such a Darwinian process hypothetically plays out *both* between proliferating, sexually reproducing biological organisms *and* fast-proliferating states of the mind-brain of individual organisms across Everett branches. Thanks to hundreds of millions of years of natural selection, the most dynamically stable phenomenally-bound system-environment correlations are the non-psychotically bound phenomenal objects populating our waking world-simulations. Psychotic binding in maladapted organisms does still occur, *comparatively* infrequently; but statistically, one's waking consciousness (as now) is overwhelmingly likely to consist in non-psychotically bound states of an adapted organism, not the Earthly counterpart of a Boltzmann brain. What we're calling "informational" and "psychotic" binding should be conceived dimensionally rather than categorically. Thus a fleeting quantum-coherent superposition of distributed neuronal feature-processors experienced as, say, a flying purple dragon is psychotic in the context of the ancestral environment of adaptation, whereas fleeting quantum-coherent neuronal superpositions of distributed feature-processors experienced as an approaching lion were potentially hugely fitness-enhancing in the extra-neural presence of a hungry predator. But flying purple dragon superpositions are not intrinsically psychotic, any more than the phenomenally-bound features of predatory lion superpositions are inherently referential – on pain of a magical theory of reference. Indeed, in some future fantastical techno-utopia – or immersive VR with different laws from basement reality – flying purple robo-dragon superpositions could be functionally non-psychotic. They might track patterns in the local mind-independent environment. What counts as sanity is contextual.

For illustrative purposes, an example with somewhat greater ecological validity than neuronal flying purple dragon superpositions might be in order. Imagine a savannah-dwelling herbivore with two disorders of phenomenal binding: *both* simultanagnosia and cerebral akinetopsia ("motion blindness"). Not merely can the herbivore's doubly unbound mind apprehend only a single perceptual object at a time; the object's progressive motion can't be perceived. So not merely is just a single member of an approaching pride of hungry lions apprehended within the herbivore's CNS world-simulation; the hungry carnivore in question just appears successively nearer without perceptibly advancing. Such a neurologically devastating condition might seem a sure-fire recipe for the hapless herbivore becoming lunch. Today, such a grisly fate would be almost inevitable. Yet to survive and genetically propagate, the doubly-unbound ancestral herbivore doesn't need to outrun the approaching lions – merely to run faster than other members of the herd. If his or her conspecifics are capable only of psychotic binding – or if their neurons are merely effectively classical or phase-scrambled neuronal "mind-dust" – then our doubly mentally unbound herbivore actually has an immense selective advantage

over every other member of the herd. For even weak and partial non-psychotic phenomenal binding confers a huge selective advantage over organisms that lack non-psychotic binding (at anything above chance levels) altogether. Or to use another, evolutionarily more ancient example, imagine a simple organism with a heritable predisposition to apprehend phenomenal patches of darkness and light – as distinct from the heritable predisposition of its conspecifics to instantiate merely discrete, decohered, effectively classical dark or light neuronal "pixels". This primordial proto-binder can functionally distinguish night from day, and safely graze (or filter-feed) rather than burrow to safety as needed in the shadow of a looming predator. Such an adaptation would be powerfully fitness-enhancing. Over evolutionary history, non-psychotic binders would outcompete psychotic binders, and superbinders will outcompete binders, culminating in the currently supreme superbinder of them all, *Homo sapiens*.

Note that on this account, Darwinian selection pressure plays out *both* between proliferating, sexually reproducing organisms across the generations and also between ultrafast-proliferating neuronal superpositions of the CNS. For although (we conjecture) next-generation matter-wave interferometry will robustly detect a perfect structural match between the reported bound phenomenology of our minds and non-psychotic neuronal superpositions, nonetheless post-Everett QM suggests that fleeting, erratic, nonsensical superpositions really do exist; they are merely of vanishingly rare measure *compared to* the information-bearing superpositions favoured by natural selection. Thankfully, experimental interferometry rather than speculative philosophising will decide the issue.

8. A Mendeleev Table for Qualia?

If sentient agents are to understand the intrinsic subjective properties of matter and energy, or to map out what we naively call the "neural correlates of consciousness", or most ambitiously, to devise a comprehensive "Mendeleev table" for qualia, then the diverse subjective textures of consciousness will play an inescapable role in the investigation by the very nature of the task. Intelligent agents will need to re-engineer themselves – genetically, pharmacologically, neurologically – in order to *instantiate* the subjective physical states in question. We'll need to become a full-spectrum "super-Mary"⁽⁴⁴⁾, so to speak – investigating state-spaces of consciousness disclosed by configurations of matter and energy that have never before been recruited for any information-processing purpose. Such state-spaces of consciousness are currently beyond the scope of scientific investigation.

By contrast, classical digital zombies cannot explore the nature of sentience; their circuitry wouldn't understand what they were investigating, let alone be cognisant of its mechanisms. This far-reaching task falls to bound phenomenal minds. A combinatorial explosion of possibilities means that the investigation of the alien state-spaces of consciousness may take millions of years, perhaps billions or more. By contrast, constructing the mathematical formalism of a unified TOE over the next

few decades may prove surprisingly easy. [Just email the author for details.]

Early in the twenty-first century, we commonly assume that physical scientists research the objective properties of matter and energy. This is true – up to a point. If physicalistic idealism is correct, then this commonplace is no more than a half-truth. For the intrinsic, subjective, first-person properties of matter and energy are real, objective and amenable to formal description via the evolution of the universal wavefunction, just as are the third-person relational properties – the properties captured by the formalism of relativistic quantum field theory or its successor. In short: we've mastered the right formalism, just assumed the wrong materialistic ontology. Subjective experience and phenomenal binding are a Hard Problem for the classical scientific materialist in the same way that fossils are a Hard Problem for the Creationist. In both cases, the anomaly in question demands a major revision of the believer's conceptual scheme. In both cases, believers are prone to spending their lives in denial.

On the face of it, to pronounce on the nature of what physical science is actually investigating might seem presumptuous for anyone but a professional physicist. Yet we don't allow the fact that, say, Newton believed he was investigating divine mechanical clockwork, or that he fancied his foremost achievement was his interpretation of the Book of Daniel, to impugn Newton's status as the greatest scientist who ever lived. Likewise, it's no disrespect to the greats of contemporary mathematical or experimental physics to say that we still don't understand the intrinsic nature of physical reality. Likewise, it's no disrespect to hard-working neuroscientists to say that we simply don't understand the mind-brain when its defining feature, consciousness, is physically impossible within the reigning materialist paradigm of science.

In a similar vein, to assert that mathematics investigates patterns of quantity, structure, space, and change would seem a commonplace. The claim that maths is really about *qualia*-patterns sounds bizarre. More telling is Bertrand Russell's jaundiced observation "Mathematics may be defined as the subject in which we never know what we are talking about, nor whether what we are saying is true." If idealistic physicalism is correct, then mathematics is ultimately about computable patterns of qualia: their quantity, structure, and change. Once again, perhaps we've mastered the formalism rather than adequately grasped the underlying ontology whose relations it captures.

9. Towards A Post-Galilean Science of Mind.

"If a potato or rutabaga can utilize quantum coherence, it's likely our brains could have figured it out as well."
(Jack Tuszynski of the University of Alberta)

A comprehensive account of reality entails an understanding of the first-person and third-person properties of the natural world – and the mathematically formalised interrelationships between them. If the distinction between the first-person and third-person properties of matter and energy were completely clean, as assumed by traditional AI, then the causal capacity of

cognitive agents to allude to both the subjective and formal properties of mind would be physically impossible in the first instance. Thus an insentient p-zombie would be physically unable, for example, to refer indexically to *this* particular self-intimating thought, or to investigate the nature of phenomenal binding, or to explore the nature of the "fire" in the equations that is responsible for the existence of sentient minds for non-zombies to describe. For a notional materialist p-zombie, it isn't even "all dark inside".

The necessity of the experimental method in scientific investigation of the third-person properties of matter and energy has been recognised since Galileo. The intellectual achievements of physical science, as traditionally conceived, are widely celebrated. By contrast, experimental investigation of the great majority of intrinsic, first-person properties of matter and energy is stigmatised and even criminalised. States of sentience as different as waking from dreaming consciousness are outlawed. Instead of Nobel laureates, research grants and lavish institutional funding, an empirically-driven exploration of the first-person properties of matter and energy plays out mainly within the scientific counterculture. An entire realm of drug-catalysed knowledge is proscribed as somehow cognitively illegitimate.

Human ignorance is unlikely to last indefinitely. If intelligent agents are to understand the natural world, then the methodology pioneered by Alexander "Sasha" Shulgin (1925-2014) in "PiHKAL"⁽⁴⁵⁾ must be integrated with mainstream academic science: an authentically post-Galilean science of physical consciousness.

Does the claim that biological agents – and perhaps mature nonbiological quantum computers centuries hence – can solve problems too difficult for a classical system to pose or answer violate the Church-Turing thesis⁽⁴⁶⁾, i.e. that any effective computation can be carried out by a Turing machine? By itself, technically, no. After all, a notional classical digital computer could be programmed to code the chemical base-pairs for the genotypes of biological super-Shulgins whose phenomenally bound minds could then explore the manifold varieties of sentience and map out the psychophysical relationships between them. Yet such a whimsical proposal doesn't mean that a classical digital computer could itself ever support a unitary full-spectrum (super)intelligence. Non-classical phenomenal binding is a necessary precondition for full-spectrum general intelligence. For without phenomenal binding, there is no unitary agent who is (un)intelligent in the first instance, let alone a general problem-solver who can systematically investigate the first-person and third-person properties of the physical world.

What is sorely lacking here is a rigorous account of computation that can handle the investigation of myriad state-spaces of qualia as well as the traditional staples of third-person computing. This challenge doesn't count as a well-defined or even meaningful question within the reigning paradigm of computer science. Sentient organic minds are biological devices that can answer questions beyond those a classical Turing machine can answer *or*

even pose – not because we are "oracles" but because – if the conjecture outlined here is experimentally vindicated – we are sentient, phenomenally bound quantum computers. Full-spectrum superintelligence will entail a seamless mastery of both the formal and the subjective properties of mind: the creation of a mature civilisation of super-Shulgins-cum-super-Turings. Recursively self-improving organic robots are poised to modify their own source code⁽⁴⁷⁾ and bootstrap our way to full-spectrum superintelligence. How closely posthuman conceptions of the physical resemble anything humans would recognise⁽⁴⁸⁾ is an open question.

10. Summary and Prospects.

The Hard Problem of Consciousness Solved; the Explanatory Gap Closed; the Binding Problem Tamed; Zombies Banished; and Physicalism Saved.

Let's recap. Here are our key assumptions and the weird but experimentally falsifiable prediction that follows. If the prediction fails, then our defence of idealistic physicalism is refuted.

- 1)** Strong emergence is false. Physicalism is true. No "element of reality" is missing from the equations of tomorrow's physics and their solutions.
- 2)** Consciousness discloses the intrinsic nature of the physical. Therefore, rudimentary consciousness occurs, not just at ultra-small distance scales, but also at ultra-short time scales. A future Planck-scale unification of quantum gravity will presumably capture the ultimate "psychon" of Planck-regime consciousness.
- 3)** The unmodified, unsupplemented formalism of post-Everett quantum mechanics is correct. "Hidden variables", Bohmian mechanics, and dynamical collapse theories of wavefunction collapse are false. Thus macroscopic quantum-coherent neuronal superpositions occur in the mind-brain. At sufficiently fine-grained temporal resolutions, the entire mind-brain exists in a single, conscious, quantum-coherent superposition. A succession of ultra-rapidly decohering virtual world superpositions constitutes biological minds. Internally, world-simulations typically seem classical. Their vehicles, i.e. phenomenally bound organic minds, are irreducibly non-classical.
- 4)** Direct realism about perception – and hence the notion that neurosurgeons or experimenters ever directly "observe" anyone else's decohered classical brain or decohered classical neurons – is false. When notionally "observing" our surroundings, both awake and dreaming organic minds instantiate individual bound perceptual objects ("local" neuronal binding) that populate dynamic world-simulations undergone by a fleetingly unitary phenomenal self ("global" binding). Phenomenal binding is not a classical phenomenon. Instead, phenomenally bound quantum-coherent neuronal superpositions have been recruited by natural selection to generate seemingly mind-independent, ostensibly classical virtual worlds. When awake, quantum biocomputers generate such pseudo-classical worlds to track fitness-relevant patterns in our local environment. Except in a dreamless sleep or coma, organic mind-brains are not decohered "pixels" of discrete neuronal micro-experiences.

The Retrodiction.

We are not zombies. Nor are we quasi-zombies, i.e. patterns of decohered neuronal "mind-dust". So there is no Hard Problem of consciousness and, in principle, no binding problem either: we're not micro-experiential zombies. Instead, we are fleetingly unitary phenomenal minds. Empirical evidence that our minds are quantum computers lies in front of our (virtual) eyes.

The Novel, Experimentally Falsifiable Prediction.

Next-generation interferometry will detect the sub-femtosecond signature of quantum-coherent neuronal superpositions in the mind-brain in the guise of quantum interference effects AND these indirectly detected quantum-coherent neuronal superpositions will robustly implicate all and only the synchronously firing feature-mediating neurons that orthodox neuroscience suggests are activated when individual phenomenally bound objects are perceived by the experimental subject.

Both predictions must be experimentally borne out in order to vindicate the quantum mind-binding conjecture outlined here. So if either no neuronal superpositions are detected, i.e. if the unitary evolution of the state vector breaks down in the mind-brain, OR if their interference signature is indeed deciphered but also implicates neurons irrelevant to the neuronal feature-mediators of the particular phenomenally bound object(s) that the experimental subject or trained up *in vitro* neuronal network reports seeing, i.e. if the interference effects detected are functionally just molecular "noise", then our quantum mind conjecture will be falsified. Falsified too would be our attempt to save physicalism.

Experimentally detecting – or definitively failing to detect – the nonclassical interference effects diagnostic of *local* phenomenal binding in the CNS will be technically less challenging than detecting the predicted trans-cerebral quantum interference effects diagnostic of *global* phenomenal binding and hence the unitary phenomenal self of everyday experience. Yet the quantum mind-binding conjecture will – provisionally – be vindicated if the signature of even local neuronal superpositions in their predicted guise are found. By analogy, if a bizarre but nonetheless falsifiable conjecture predicts (what orthodox neuroscience would claim is) the equivalent of little green pixies living at the bottom of the garden, and – amazingly – a single little green pixie is unequivocally detected, then we wouldn't withhold assent to the bizarre conjecture on the grounds that experiment hadn't yet detected the theorised pixie breeding colony.

Further Challenges.

1) The mechanisms supporting the succession of differentially robust sub-femtosecond neuronal superpositions that – hypothetically – underpin phenomenal binding must be elucidated at the molecular level. Only at the molecular level can philosophical hand-waving be turned into real, measurable, quantitatively exact physical science. At much longer time-scales of milliseconds and above, the standard coarse-grained story from connectionist neuroscience and dynamical systems theory takes over from the femto-mind regime. Thus whether we are in a dreamless sleep, dreaming or wide awake, our memories are coarsely encoded in the connectivity, connection weights, and the internal

architecture of our neurons after our neural networks have been progressively "trained up". Besides its idealist ontology, the quantum mind-binding conjecture explored here to save physicalism from the spectre of Chalmersian dualism is radically unorthodox only insofar as what mainstream neuroscience reckons is the mere synchronous firing of classical neuronal distributed feature-processors is conjectured instead to be a succession of quantum-coherent neuronal superpositions. Only experiment can corroborate or falsify this hypothesis. If the prediction fails, then our defence of idealistic physicalism is refuted too.

2) Even if non-materialist physicalism is true, the lack of some sort of Rosetta Stone to "read off" the values of qualia – both bound and unbound – from the solutions to the field-theoretic equations of QFT is a huge challenge. Compare a much more straightforward identification. Nowhere in Maxwell's field equations is light explicitly identified with electromagnetic radiation. But once the value of the constant c was calculated – around 300,000 kilometres per second – then the identity of its value with the known velocity of light made the identification inevitable. In other words, no "element of reality" was missing from Maxwell's formalism, or, more strictly, from its subsequent quantum electrodynamic generalisation. Likewise, if idealistic physicalism is true, no "element of reality" is missing from the formalism of relativistic quantum field theory or its currently speculative successor. However, in contrast to the ease of identification of light with visible frequencies of electromagnetic radiation, the conjecture that the solutions to the equations of QFT yield the precise values of all and only physically possible experiences amounts to both a mathematical straitjacket and a veritable Pandora's box. For the only way cognitively to grasp the values of the diverse subjective properties of the physical fields of experience that the solutions to the formalism encode is personally to instantiate bound neuronal superpositions of these subjective properties. Even after extensive psychotropic and eventually neurogenetic experimentation, myriad forms of consciousness will presumably be forever inaccessible to rational mind – though equally, many physical systems that today we might naively imagine could in future be unitary subjects of experience, notably ultra-powerful classical digital computers or nonbiological classical connectionist systems, will always be effectively insentient.

3) Whether our conscious minds are essentially classically parallel, connectionist systems, or quantum supercomputers as conjectured here, another enigma remains. The late evolutionary neurological mechanism by which a massively parallel biological neurocomputer generates a virtual classical machine – the slow serial stream of one's logico-linguistic thinking via which this paper is written and read – is unknown. We do know of crude methods to disrupt our stream of logico-linguistic thought-processing. For example, taking LSD induces the "flooding" phenomenon that disrupts serial thought, whereas low-dose psychostimulants tend modestly to enhance logico-linguistic thought. Yet that's as far as it goes. Whatever the nature of this virtual seriality-generating mechanism in the CNS, we can sketch out an evolutionary chronology of information processing

systems. An irreducibly quantum multiverse first generated information-bearing self-replicators – biological life – which manufactured quantum supercomputers in the form of central nervous systems, one species of which spawned the serial, logico-linguistic virtual machines currently unique to human minds. These serial virtual machines conceived and created classical digital computers, then classically parallel artificial connectionist systems, and finally – though here we run a little ahead of our story – artificial nonbiological quantum computers. The long-term interplay of these multiple architectures is hard to foresee with any confidence; but the destiny of sentient life in the cosmos most probably lies in full-spectrum superintelligence⁽⁴⁹⁾.

* * *

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[David Pearce](#)

e-mail: dave@hedweb.com

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Notes

Appendix

Physicalist Hotlinks

Idealistic Physicalism (mp3)



Talks

BLTC Research

Superhappiness?

Quantum Ethics?

The Abolitionist Project

Physicalism (Wikipedia)

Physicalism (Grokopedia)

What is a Quantum Mind?

The Biointelligence Explosion

Social Media Postings (2026)

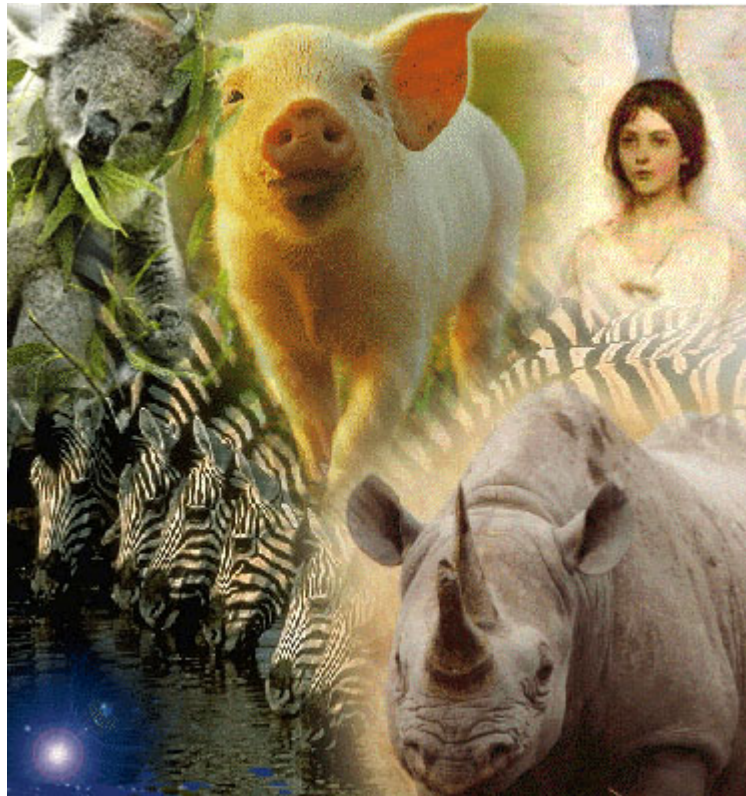
MDMA: Utopian Pharmacology

Full-Spectrum Superintelligence

What is Non-Materialist Physicalism?

ChatGPT on Non-Materialist Physicalism

Quora Answers on Physicalism etc (2015-25)



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